

1(a)	any two from: 1 (by) contracts / contracting / contraction, and, relaxes / relaxing / relaxation ; <i>must be in context of bronchioles</i> R if includes <i>ref. to stretch / expand / recoil</i> R if in context of <i>bronchus or trachea (then ecf in subsequent mps)</i> 2 diameter / lumen size, can be, controlled / changed ; AW A contraction, decreases diameter (of lumen) / constricts A relaxation (after contraction) increases diameter (of lumen) / widens / dilates / expands R vasoconstriction / vasodilation 3 <i>idea of</i>, controls / regulates, flow of air / volume of air flowing ; AW I statements such as 'lets air in' / 'allows inhalation and exhalation' A oxygen in / carbon dioxide out, <i>instead of air flow</i> R oxygen out and carbon dioxide 4 AVP ; e.g. contract to, reduce air movement / prevent entry of contaminants <i>idea of</i> normally relaxed but can, contract / cause constriction, when needed <i>max 1 if response relates function to inhalation and exhalation</i> <i>0 marks if incorrectly stated as <u>elastic tissue or elastic fibres</u></i>	2
1(b)(i)	stage micrometer / stage micrometer scale ;	1

Question	Answer	Marks
1(b)(ii)	yes, qualified ; e.g. 0.2 mm = <u>200 μm</u> 0.2 mm / smallest student can see, is less than <u>0.25 mm</u> 250 μm = <u>0.25 mm</u> 250 μm / cell length, is longer than <u>200 μm</u> the cell is <u>50 μm</u> longer (than 0.2 mm) the cell is <u>0.05 mm</u> longer (than 0.2 mm) <i>allow if converted to standard form (2×10^{-4} v 2.5×10^{-4} m)</i> <i>must have units</i> <i>must include numerical values</i>	1
1(c)(i)	any two from: wall, of one layer / is one cell thick ; 1 capillary is one cell thick A thin wall / endothelial cells are a single layer A endothelium is, a single layer (of cells) / one cell thick <i>idea that</i> red blood cells transported in single file ; <i>idea that</i> red blood cells diameter (approximately), as large / same, as capillary (lumen) ; A if diameter (e.g. 7μm) stated as being similar to capillary size <i>idea of</i> diameter of capillary narrower than smooth muscle cell ; AVP ; e.g. no muscle and elastic layer muscle fibres / elastic fibres / collagen, not present <i>ref. to</i> different shapes of red blood cell / red blood cell showing flexibility, to pass through (narrow) capillary	2

Question	Answer	Marks
1(c)(ii)	<i>allow tissue fluid / fluid surrounding cells, instead of (smooth muscle) cells</i> <i>allow, blood / plasma, for capillary</i> <i>allow O₂ for oxygen and CO₂ for carbon dioxide</i> any three from: <i>capillary function</i> 1 named, substance / type of substance, that is supplied to smooth muscle cells or is a product removed from the cells ; e.g. (supply) oxygen / nutrients / glucose / amino acids / cell-signalling molecule e.g. (removes) carbon dioxide / lactic acid / waste A exchange of respiratory gases (between capillary and cells) I plasma proteins into tissue fluid <i>structure to function</i> 2 thin wall / AW, so short distance ; R capillary is thin 3 endothelial pores / fenestrations, for more efficient passage / AW ; A gaps, between / within, (endothelial) cells (of capillary wall) A increases, quantity / rate of supply (of substances) A endothelial pores for formation of tissue fluid (around cells) R if <u>blood</u> leaves to make tissue fluid 4 (endothelial) pores / AW, for passage of, phagocytes / monocytes / macrophages / neutrophils ; 5 small (size) / narrow diameter, to reach cells / so (all) cells are close ; A value in range 5–10 µm A small size / narrow lumen, slows (rate of) blood flow 6 detail about <u>red blood cell</u> and, respiratory gases / gas exchange / oxygen / carbon dioxide ; e.g. narrow lumen slows red blood cell movement for uptake of oxygen slows red blood cell movement to increase time for exchange of gases only one red blood cell fits (at a time) so short diffusion distance red blood cells are close to (muscle) cells for gas exchange red blood cells squeeze through, (so) distance for oxygen to reach cells is minimised	3

Question	Answer	Marks
1(d)(i)	(a gene mutation is a) change in the sequence of base pairs in a DNA molecule ; A change in the DNA, base / nucleotide, sequence I change in RNA base sequence I names of mutations codes for / may result in / results in / AW, an altered / a different / a changed, polypeptide ; A protein / amino acid sequence / primary structure, <i>for polypeptide</i>	2
1(d)(ii)	<i>any two from:</i> differences owing to, RNA / gene, splicing ; A primary transcript for RNA A <i>ref. to</i> alternative splicing R DNA splicing R mutation occurring during gene splicing (differences occur during) removal of, introns / non-coding sequences ; R if introns described as bases / nucleotides / codons R different number of introns removed / some introns remain exons / coding sequences, detail ; e.g. joined differently / in a different combination in middle (of transcript), removed / not included for repeating sequence, removed / not included (so) messenger RNA / mRNA, formed is different ;	2

Question	Answer	Marks
1(d)(iii)	<i>allow ref. to binding sites as plural</i> <i>I ref. to active site(s)</i> <i>any one from:</i> (still have) similar, tertiary structure / binding site shape ; A similar 3D shape R same A <i>idea of</i> binding sites still complementary to attach to (the) proteins removed amino acids are not structural amino acids ; removed amino acids are not part of binding site ; A description R-group interactions (still) the, same / similar ; AVP ; e.g. idea of more than one binding site and only, one / a few, changed A same type of binding site, but a different number	1